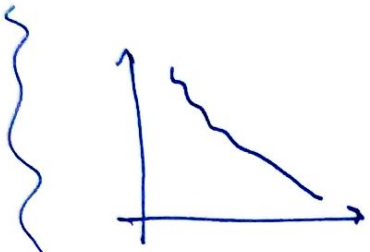
F-BLOCK →

① Only one stable oxid^{n} state → **+3** → yehi mostly ques mein hoga.

LATHANOIDS →

① Lanthanoid Contraction



Imperfect shielding one e^- by another in the same subshell

→ Consequence ⇒ Zr ≃ Hf (160pm) (159 pm), Nb ≃ Ta, Mo ≃ W

② $Ce^{+4}/Ce^{+3} = +1.74\ V$

→ Ce(IV) good analytical reagent so reach is slow

$Eu^{+2} \longrightarrow Eu^{+3}$ $\quad Tb^{+4} \longrightarrow Tb^{+3}$

$Yb^{+2} \longrightarrow Yb^{+3}$

* Ln^{+3} most stable inspite of its e^- confi.

③ La

Celina aur Preety ne dande se Pammy aur Simmy ko maara

Ce, Pr, Nd, Pm, Sm

Europe Gya Tab diarrhea ho gaya

Eu, Gd, Tb, Dy, HO

Errey! Tamatar yellow aur blue hai.

Er, Tm, Yb, Lu

④ Silvery white soft metals

Samarium → steel hard

Good conductors of heat and electricity

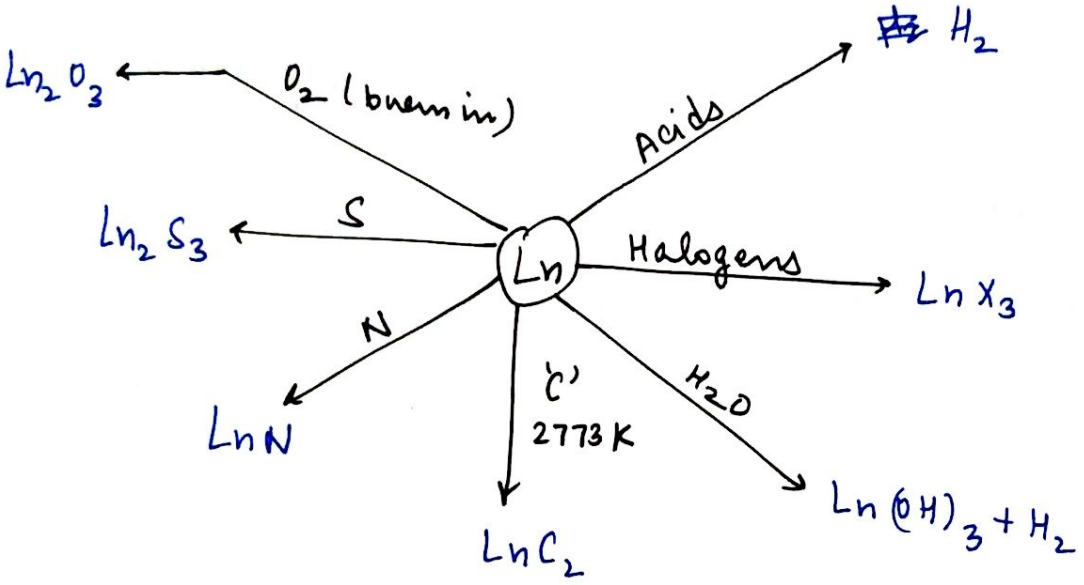
⑤ La^{+3}, Lu^{+3} → no colour
Rest colour ✓

⑥ La^{+3}, Ce^{+4}, Yb^{+2}, Lu^{+3}] → All diamagnetic
(La^{+3}, Ce^{+4} → f^0; Yb^{+2}, Lu^{+3} → f^{14})

⑦ Exchange Enthalpy impacts stab.

$Ln^{+3} + 3e^- \rightarrow Ln$ ($-2.4V < E_{SRP} < -2.2V$)

For Eu → $-2V$



(Ln_3C, Ln_2C_3 are also formed)

⑧ Alloy → Misch metall → Lanthanoid metal (95.1.) → Used in Mg-based alloy for bullets, shell, flint etc.
→ Iron (5.1.)

Ln oxides → Television screens, flouresecing surfaces

⑨ Exceptions in Confi. →

$La \rightarrow 5d^1 6s^2$

$Ce \rightarrow 4f^1 5d^1 6s^2$

$Gd \rightarrow 4f^7 5d^1 6s^2$

⑩ Ce^{+4} is well known to exhibit +4 O.S.

⑪ $La(OH)_3 \longrightarrow Lu(OH)_3$
Basicity decreases

⑫ Difficulty in sep. of 4d, 5d elements as they occur in nature together, as of similar radii

ACTINOIDS →

AC

① Tnode Pehlwan Unse Niptenge
Th Pa U Np

Pueane Aam Cum Bikenge
Pu Am Cm Bk

Computer Engeneering
Cf Es

Fakeer Mia ne Li Rakam
Fm Md No Lr

② Exceptions in Confi. →

Ac → $6d^1 7s^2$

Th → $6d^2 7s^2$

Cm → $5f^7 6d^1 7s^2$

Pa → $5f^2 6d^1 7s^2$
U → $5f^3 6d^1 7s^2$
Np → $5f^4 6d^1 7s^2$
→ d^1

③ Actinoids → Radioactive

④ 5f e^- can participate in bonding more than 4f e⁻s as they are not buried as 4f orbitals.

⑤ Actinoid Contraction



⑥ General O.S '+3' only

Th → +4 (Exception)

Np, Pu → most no. of O.States (3, 4, 5, 6, 7)

(7) i) Silvery in appearance

ii) Display variety of structures

iii) Highly Reactive

iv) Al ki trh protective Oxide form with HNO_3

v) Actinoids ki magnetic prop. more complex than Lanthanoids

vi) IE of actino. < Early Lantha.

(8) Acti. contra. greater from element to element than Lanth. Contra. as of poor shielding of 5f.

(9) i) TiO → Pigment Industry

ii) MnO_2 → Dry battery cells
↳ also req. Zn and Ni / Cd

iii) Grp-11 → Coinage Metals

iv) Silver → UK coins → Cu/Ni alloy

v) V_2O_5 → catalyst

vi) $TiCl_4$ and $Al(CH_3)_3$ → Zeigler Nata Catalyst (Polyethylene) (Polythene)

vii) $H-C\equiv C-H \xrightarrow{PdCl_2} CH_3CHO$ (WACKER PROCESS)

viii) Photographic Industry → light sensitive prop. of AgBr